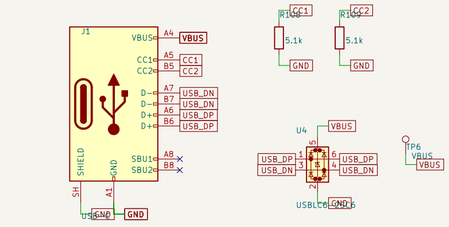
Lira, wired: Raspberry Pi RP2040, USB-C only (no battery), 57 keys, built-in USB sound card

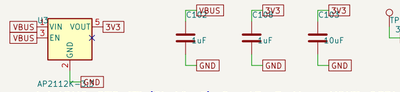
The wireless board (branch main: nRF52840 module, Li-Po) with the radio and the battery taken out; the RP2040 follows Raspberry Pi's 'Hardware design with RP2040' (the Pico's circuit). Differences from the wireless board:

1. USB only, no battery: no Ebyte E73 (nRF52840) module, no Bluetooth, no Li-Po cell, no charger (MCP73831), no power path (BS819W + Si2301), no power switch, no battery divider, no 32.768 kHz crystal, no antenna keep-out.
2. Raspberry Pi RP2040 instead of the module, with the minimum circuit of Raspberry Pi's 'Hardware design with RP2040' (the Pico's): 12 MHz crystal ABM8-272-T3 (the one Raspberry Pi qualified) with 2 x 15 pF and 1k in series on XOUT; W25Q128JV QSPI flash (16 MB) with a 10k pull-up on QSPI_SS; 100 nF on each of the six IOVDD pins, on both DVDD, USB_VDD and ADC_AVDD, 1 uF on VREG_VIN and VREG_VOUT; the internal regulator makes the 1.1 V core supply; TESTEN grounded.
3. USB: hub port 3 goes to the RP2040's USB_DP / USB_DM through 27 ohm series resistors (as on the Pico and in the design guide), instead of the module's D+ / D-.
4. Programming: the RP2040's boot ROM is a USB drive (RPI-RP2) - no bootloader to flash with an SWD probe. A blank board starts in it; BOOT (SW101, QSPI_SS to GND through 1k) held while plugging the cable enters it again; RESET (SW102) pulls RUN low. The SWD pads stay for debugging.
5. 3.3 V regulator AP2112K-3.3 (the same as the wireless board) fed straight from VBUS: 1 uF in, 1 uF + 10 uF out.
6. Status LED (red, Caps Lock) on an RP2040 GPIO, through 1k, as on the wireless board.
7. Test points: SWD (SWCLK, SWDIO, RESET, GND) in one row in front of the RP2040, 3V3 behind it, VBUS; no VBAT / VIN.
8. Sound card: the supply switch U8 is turned on by VBUS instead of a GPIO (the wireless board has it on only in USB mode; here it always is) and its pull-down is dropped. It still soft-starts the sound card's 5 V: its 17 uF are charged through the switch's controlled rise, not straight from the cable (only ~3 uF sit directly on VBUS). The amplifier's EN delay (1M + 470 nF) runs from VBUS too, and D104 drags EN down with VBUS on unplug.

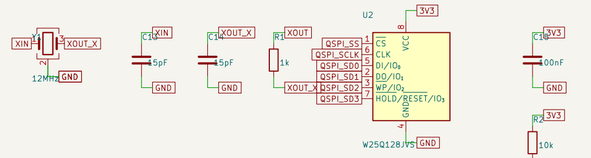
USB-C (sink: 5.1k on each CC) and ESD protection



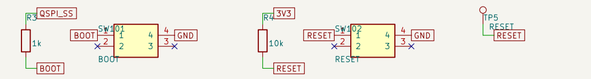
3.3 V regulator AP2112K from VBUS (EN tied to VIN); 10 uF bulk at the RP2040



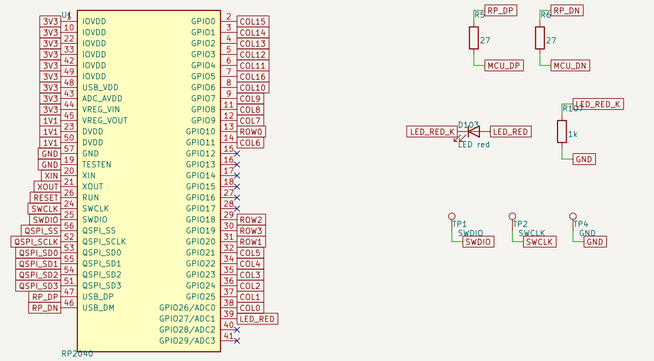
12 MHz crystal ABM8-272-T3 (CL 10 pF): 2 x 15 pF, 1k on XOUT; QSPI flash W25Q128JV, /CS pulled up



BOOT: QSPI_SS to GND through 1k (USB boot ROM); RESET: RUN to GND, 10k pull-up



Raspberry Pi RP2040 (1.1 V core from its own regulator: VREG_VOUT -> DVDD); USB through 27 ohm to the hub's port 3; status LED (Caps Lock) on a GPIO; SWD pads

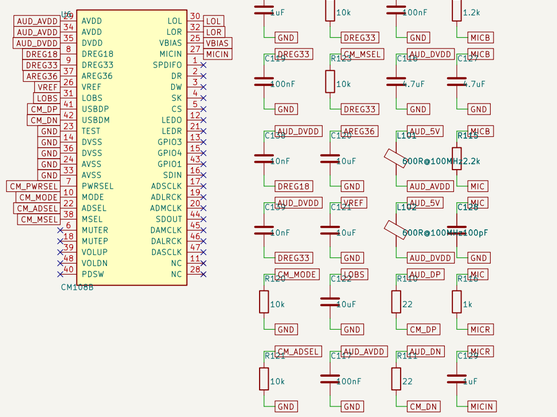


Supplies: VBUS from the receptacle, codec supplies from VBUS through ferrite beads



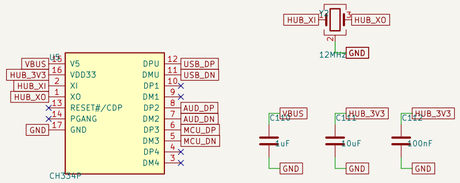
Sound card CM108B (C-Media demo board v1.33 sheet 4, app notes v1.50); headset jack, CTIA

Open pins: internal pulls select headset mode (MODE low), 100 mA bus power (PWRSEL high), internal ADC (ADSEL low), mixer on (MSEL high); R120-R123 strap them (10k). Jack: tip L, ring 1 R, ring 2 GND, sleeve mic.



USB hub CH334P (datasheet V2.91)

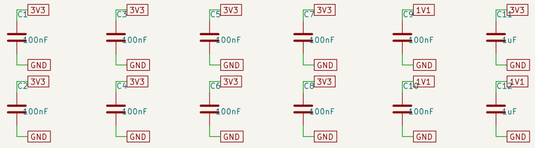
upstream from the receptacle, port 3 -> RP2040, port 2 -> codec



12 MHz crystal without load capacitors (inside the CH334)

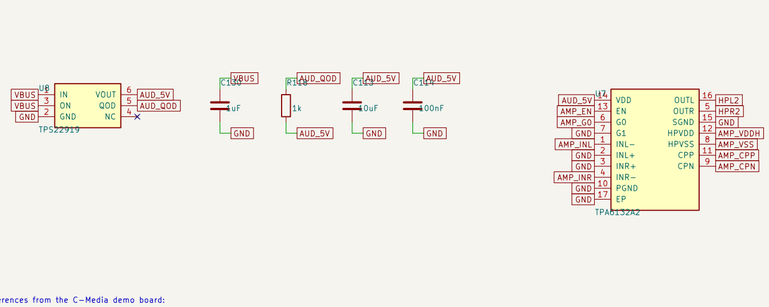
RESET#, PGANG and ports 1 / 4 left open

RP2040 decoupling: 100 nF at each IOVDD, USB_VDD, ADC_AVDD and DVDD pin; 1 uF at VREG_VIN and VREG_VOUT



Key matrix, col2row: column -> switch -> diode (cathode on the row). Labels: (row,col) of each key.

Sound card supply: TPS22919 load switch, turned on by VBUS (soft start); headphone amplifier TPA6132A2 (DirectPath, 0 dB; -6 dB with JP3 closed); headset jack, CTIA



Differences from the C-Media demo board:

1. Supply: VBUS through a TPS22919 load switch (soft start 1.3 ms, output discharge) to AUD_5V, ON tied to VBUS (the wireless board drives it from a GPIO, on only in USB mode): the sound card's capacitors are charged slowly after the cable has made contact. A hard switch-on would dip VBUS enough to reset the hub (CH334 datasheet 6.1).
2. Headphones: a TPA6132A2 DirectPath amplifier after LOL / LOR (demo board: 100 uF + 39R straight to the jack): about 0 ohm output impedance, no coupling capacitors, 0 dB gain (-6 dB with JP3 closed, for very sensitive earphones). Inputs single-ended through 1 uF (8 Hz); EN about 0.15 s after the supply (1M + 470 nF from VBUS), and dragged down with VBUS through a Schottky when the cable is pulled (EN stays within a Schottky drop of VBUS). On unplug the codec and the amplifier lose their supply together: listen for a click.
3. Microphone: bias 1.2k + 2.2k (demo board 2.2k + 8.2k, for PC microphones) for phone headsets (Android: 1.8-2.9 V through 2.2k); 1k in series before MICIN against the step when a plug goes in.
4. ~~The demo board's two 10 nF capacitors DVDD - DREG18 and DVDD - DREG33 ('close to IC', for ESD) are fitted.~~
5. The amplifier's SGND (pin 15) is grounded at the amplifier (its ground pad and a via into the top ground plane), not on a track of its own to the jack's ground pin as TI suggests (datasheet 10.3): with both layers poured the difference is a few millivolts.
6. The four mode jumpers are also strapped with 10k (MODE/ADSEL low, PWRSEL/MSEL high to DREG33) on top of the chip's internal pulls. D105-D107 are

generated by source/pcb/gen_schematic.py

Sheet: /
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Title: Lira (wired)

Size: A2	Date: 2026-10-17	Rev: A
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